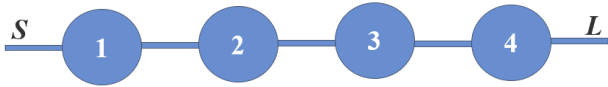


Use GA to find the coupling matrix

Filter is a communication device widely used in communication systems. The first step in designing a filter is often obtaining the coupling matrix, as shown in the following figure. Finding the coupling matrix can be considered as an optimization problem. Now, assuming you are a microwave filter engineer, you are required to obtain a coupling matrix that meets the frequency response requirements (i.e., constraints).

The filter structure is as follows:



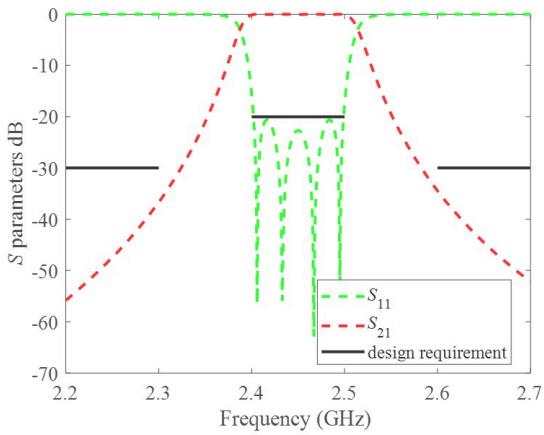
The coupling matrix is as follows:

$$\begin{bmatrix} 0 & x(1) & 0 & 0 & 0 & 0 \\ x(1) & 0 & x(2) & 0 & 0 & 0 \\ 0 & x(2) & 0 & x(3) & 0 & 0 \\ 0 & 0 & x(3) & 0 & x(2) & 0 \\ 0 & 0 & 0 & x(2) & 0 & x(1) \\ 0 & 0 & 0 & 0 & x(1) & 0 \end{bmatrix}$$

where $x(1)$, $x(2)$, $x(3)$ are coupling coefficients, which are the decision variables for the optimization.

The design requirements are as follows:

- $\max(\text{abs}(S_{11}))$ is less than -20dB in the range of 2.4 GHz-2.5 GHz
- $\max(\text{abs}(S_{21}))$ is less than -30dB in the range of 2.2 GHz-2.3 GHz
- $\max(\text{abs}(S_{21}))$ is less than -30dB in the range of 2.6 GHz-2.7 GHz

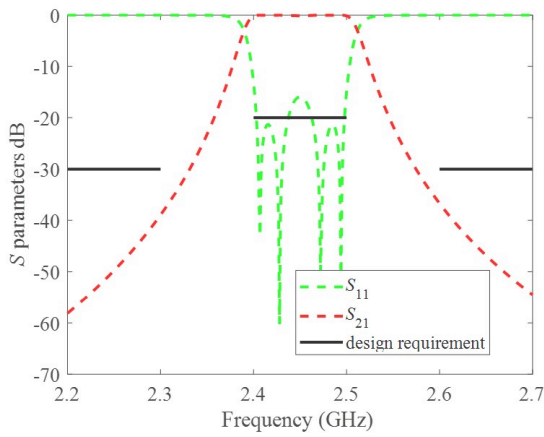


They can be considered as constraints. The sum of the constraint violations can be the objective function, which is provided. Now, your task will start.

The initial values for $[x(1), x(2), x(3)]$ are $x_0 = [0.985 \ 0.860 \ 0.650]$, and the corresponding response is shown in the following figure, which does not meet the design requirements. The task is to use the genetic algorithm (GA) to optimize $x(1)$, $x(2)$, and $x(3)$.

The objective function for coupling matrix optimization, provided in the attached file 'objective_func.m' (see reading materials), is used in the optimization. When the fitness value reaches 0, it indicates that all the requirements are met.

The lower bound for design variables is $[0.5, 0.5, 0.5]$, and the upper bound is $[1.2, 1.2, 1.2]$. The population size of GA should not exceed 20, and the number of optimization iterations should not exceed 200.



Script

 Save  Reset  MATLAB Documentation (https://www.mathworks.com/help/) [Open Item in MATLAB Online](#) 

```
1 nvars = 3 % Dimensions of design variables
2 lb = [0.5,0.5,0.5]; % Lower bound of design variables
3 ub = [1.2,1.2,1.2]; % Upper bound of design variables
4 options = optimoptions('ga', ...
5 'PlotFcn', @gaplotbestf, ...
6 'PopulationSize', 20, ...
7 'MaxGenerations', 200) % Please fill in the optimization options
8 [x, y] = ga(@objective_func, nvars, [], [], [], [], lb, ub, [], options); % Please fill in the ga function
9
10 function ff = objective_func(x)
11
12 M=zeros(5,5);
13 M(1,2)=x(1);
14 M(2,1)=x(1);
15 M(6,5)=x(1);
16 M(5,6)=x(1);
17 M(2,3)=x(2);
18 M(3,2)=x(2);
19 M(3,4)=x(2);
20 M(4,3)=x(2);
21 M(4,5)=x(2);
22 M(5,4)=x(2);
23 M(5,6)=x(2);
24 M(6,4)=x(2);
25 M(6,5)=x(2);
26 M(3,5)=x(3);
27 M(5,3)=x(3);
28 M(4,6)=x(3);
29 M(6,4)=x(3);
```

 Run Script 

Tests: All Tests Passed

Submit 

-  Is the number of generations too large? <=200
-  Is the population size too large? <= 20
-  Has GA been used?
-  Was the result correct?

Output

```
nvars =

     3

options =
```

ga options:

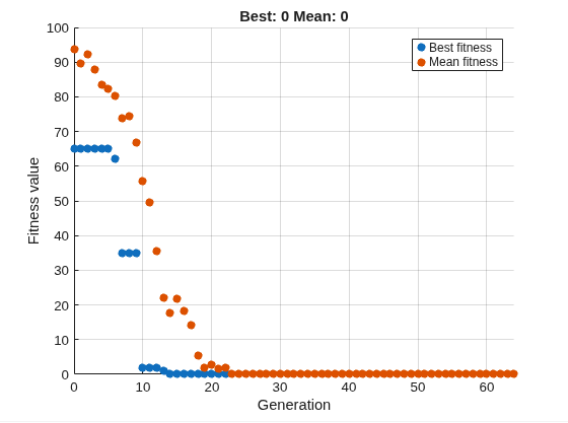
Set properties:

MaxGenerations: 200
PlotFcn: @gaplotbestf
PopulationSize: 20

Default properties:

ConstraintTolerance: 1.0000e-03
CreationFcn: []
CrossoverFcn: []
CrossoverFraction: 0.8000
Display: 'final'
EliteCount: '0.05*PopulationSize'
FitnessLimit: -Inf
FitnessScalingFcn: @fitscalingrank
FunctionTolerance: 1.0000e-06
HybridFcn: []
InitialPopulationMatrix: []
InitialPopulationRange: []
InitialScoresMatrix: []
MaxStallGenerations: 50
MaxStallTime: Inf
MaxTime: Inf
MutationFcn: []
NonlinearConstraintAlgorithm: 'auglag'
OutputFcn: []
PopulationType: 'doubleVector'
SelectionFcn: []
UseParallel: 0
UseVectorized: 0

ga stopped because the average change in the fitness value is less than options.FunctionTolerance.



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